

Sri Lanka Institute of Information Technology



Mini Project - Dataset Proposal

Fundamentals of Data Mining

IT3051

B.Sc. (Hons) in Information Technology, specializing in Data Science

Group – Northern Uni

Gowsika Surendran – IT23794566

Manoharan Subasthican – IT23810778

Kuganesan Kageepan – IT23717572

Kokulan Seyon – IT23752672

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1. Dataset Name and Original Source

1.1 Selected Dataset

The dataset proposed for this project is the Hotel Booking Demand dataset. It contains hotel reservation records for two hotel types, a Resort Hotel and a City Hotel. The dataset is appropriate for supervised machine learning because it contains booking characteristics together with the known cancellation outcome for each record.

The project will use the combined dataset distributed through Kaggle, containing 119,390 booking records and 32 columns.

1.2 Direct Dataset Source

The direct download source selected by the group is Kaggle. The dataset page is titled Hotel Booking Demand and provides the combined hotel booking dataset in CSV format. Kaggle URL: <https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>

1.3 Original Research Source

The Kaggle dataset is based on the Hotel Booking Demand datasets published by Nuno Antonio, Ana de Almeida and Luis Nunes. The associated Data in Brief publication describes booking data from one resort hotel and one city hotel and provides the academic background for the dataset.

The original research source is important for understanding the dataset context, variables and period covered by the observations.

1.4 Dataset Provenance

For this project, Kaggle will be identified as the practical download source, while the original academic publication will be cited as the research source. Clearly distinguishing these sources improves transparency and ensures that the dataset is properly attributed.

2. Dataset URL and Citation

2.1 Kaggle Dataset URL

<https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>

The Kaggle version is convenient for the academic project because the data can be downloaded in a tabular format and analysed using common data-mining and machine-learning tools.

2.2 Associated Research Publication

Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41-49. DOI: 10.1016/j.dib.2018.11.126.

2.3 Citation Approach

The final report will acknowledge both the Kaggle dataset page used to obtain the project data and the original academic publication describing the Hotel Booking Demand datasets. Any transformations, feature engineering and preprocessing performed by the group will be documented separately from the original data.

3. Dataset Description and Real-World Context

3.1 Background of the Problem

Hotel booking cancellations are a practical challenge for accommodation businesses. A reservation may be accepted weeks or months before arrival, but the customer can later cancel it. Cancellations can affect expected occupancy, room allocation, staffing, resource planning and revenue forecasting.

A data-mining system can analyse patterns in historical reservations and estimate whether a new booking resembles bookings that were previously cancelled or retained. The proposed system is intended to provide decision-support information rather than automatically make decisions about customers.

3.2 Dataset Description

The dataset contains booking-related, customer-related, stay-related and reservation-related attributes. Examples include hotel type, lead time, arrival date, length of stay, number of guests, meal type, country, market segment, distribution channel, repeated-guest status, previous cancellations, room type, deposit type, customer type, average daily rate, parking requirements and special requests.

The records cover bookings with arrival dates from July 2015 to August 2017.

3.3 Proposed Problem Scenario

The proposed problem is to predict whether a hotel booking will be cancelled or not cancelled using relevant information that is available before the final reservation outcome is known.

Prediction objective: To develop a machine-learning-based classification system that estimates the cancellation outcome of a hotel booking using relevant reservation characteristics.

3.4 Expected System Inputs

The final input set will be decided after EDA and feature selection. Potential user inputs may include:

- Hotel type
- Lead time
- Planned weekend and weekday nights
- Number of adults, children and babies
- Meal type
- Market segment and distribution channel
- Repeated-guest status and previous booking history
- Reserved room type
- Deposit type
- Customer type
- Average daily rate (ADR)
- Required car parking spaces
- Total special requests

3.5 Expected System Output

The main output will be a human-readable classification such as Likely to Cancel or Not Likely to Cancel. Where supported by the selected model, the interface may also display an estimated cancellation probability or risk level. The output will be presented as a prediction based on historical patterns, not as a guaranteed future outcome.

3.6 Proposed User Requirements

- The interface should provide clearly labelled fields for the selected booking characteristics.
- Categorical inputs should be displayed using understandable choices rather than internal encoded values.
- Numerical fields should validate missing, invalid or unreasonable input.
- The same preprocessing rules used during model training must be applied to new user inputs.
- The prediction should be displayed in simple language.
- The interface should not expose outcome-related or leakage-prone fields as user inputs.
- The system should explain that the result is decision-support information and may contain prediction errors.

3.7 Potential Users and Stakeholders

- Hotel managers
- Reservation-management staff
- Revenue-management teams
- Operational planners
- Hospitality analysts and academic users

3.8 Decisions the System Could Support

- Reservation planning
- Occupancy forecasting
- Room and resource planning
- Understanding cancellation patterns across booking groups

- Supporting revenue-management analysis

3.9 Scope and Responsible-Use Boundary

The system will identify statistical patterns in the approved historical dataset. It will not establish that any feature causes cancellation. Predictions should not be used as the sole reason to penalize, reject or unfairly treat a customer.

4. Number of Records and Relevant Features

4.1 Dataset Size and Structure

The combined dataset contains 119,390 records and 32 columns. Each row represents a hotel booking. One column, `is_canceled`, is the target variable. The remaining fields describe different aspects of the reservation or its final status.

The dataset is sufficiently large for EDA, preprocessing, model training, validation, testing and model comparison while remaining manageable for a student project.

4.2 Feature Groups

The variables can be considered in broad groups to make the dataset easier to understand.

Group	Example variables
Booking / timing	hotel, lead_time, arrival_date_year, arrival_date_month, arrival_date_week_number, arrival_date_day_of_month
Stay / guests	stays_in_weekend_nights, stays_in_week_nights, adults, children, babies
Customer / channel	country, market_segment, distribution_channel, customer_type, is_repeated_guest

History	previous_cancellations, previous_bookings_not_canceled
Reservation	reserved_room_type, assigned_room_type, booking_changes, deposit_type, days_in_waiting_list
Commercial / requests	adr, agent, company, required_car_parking_spaces, total_of_special_requests
Outcome-related	reservation_status, reservation_status_date
Target	is_canceled

4.3 Numerical Variables

Important numerical variables include lead time, stay-night counts, guest counts, previous booking counts, booking changes, days in waiting list, ADR, parking spaces and special requests. Their distributions, ranges and possible extreme values will be investigated after formal dataset approval.

4.4 Categorical Variables

Categorical variables include hotel type, meal type, country, market segment, distribution channel, reserved and assigned room types, deposit type and customer type. Suitable encoding methods will be selected according to the model and observed categories.

4.5 Variables Requiring Special Review

Some variables require careful review before modelling. reservation_status and reservation_status_date describe the eventual reservation outcome and can cause direct data leakage when predicting cancellation before the outcome is known. Other fields such as assigned room type, booking changes and waiting-list information will be evaluated against the chosen prediction point. A feature will be retained only if it is realistically available at the time the prediction is intended to be made.

5. Target Variable and Its Meaning

5.1 Target Variable

The target variable is `is_canceled`. It is binary:

- 0 = Booking was not cancelled
- 1 = Booking was cancelled

5.2 Target Distribution

The dataset contains approximately 75,166 not-cancelled bookings (about 63%) and 44,224 cancelled bookings (about 37%). The distribution is therefore moderately imbalanced but not extremely imbalanced.

5.3 Treatment of Class Imbalance

Class imbalance will be examined during EDA and model evaluation. Accuracy will not be used alone because a model can achieve a reasonable overall accuracy while performing poorly on the cancellation class. Precision, recall, F1-score, confusion matrix and ROC-AUC will also be examined.

Resampling or class-weight techniques will not be applied automatically. They will be considered only if baseline experiments show that imbalance meaningfully harms the model's ability to identify cancellations.

5.4 Interpretation

The predicted class represents the model's estimate based on patterns in historical bookings. A cancellation prediction does not mean that a customer will definitely cancel.

6. Expected Machine-Learning Task

6.1 Type of Task

The project is a supervised binary classification task because each training record has a known target label and there are two possible outcomes.

6.2 Proposed Data Splitting Strategy

After approval and initial data preparation, the labelled dataset will be divided into development and final evaluation data. A stratified split will be considered so that the cancellation proportions remain similar across subsets. A practical strategy may use separate training, validation and testing subsets, or a training/test split with cross-validation performed only on the training data. The final choice will be justified during implementation. The test set will remain isolated from model-selection and tuning decisions.

6.3 Proposed Preprocessing Workflow

- Inspect dimensions, feature names, data types and target distribution.
- Identify and treat missing values using justified methods.
- Check duplicate and invalid records.
- Investigate extreme values and outliers rather than removing them automatically.
- Encode categorical variables appropriately.
- Scale numerical variables where required by the algorithm.
- Engineer and select relevant features.
- Remove leakage-prone, outcome-related or non-useful fields.
- Fit learned preprocessing operations using training data only where appropriate.
- Use a consistent preprocessing pipeline for training and future predictions.

6.4 Proposed Feature Engineering Experiments

Feature engineering will be guided by EDA rather than fixed in advance. Two simple and interpretable candidate features are:

- Total Stay = stays_in_weekend_nights + stays_in_week_nights.

- Total Guests = adults + children + babies.

Controlled experiments can compare model performance with and without engineered features while keeping the data split, preprocessing principles, model family and evaluation measures consistent. This will show whether the engineered representation adds useful predictive information.

6.5 Proposed Baseline and Candidate Models

At least four substantive classifiers will be compared:

- Logistic Regression: interpretable baseline for binary classification.
- Decision Tree: learns nonlinear decision rules and is comparatively easy to explain.
- Random Forest: ensemble of decision trees that can model complex interactions and reduce instability.
- Gradient Boosting: sequential ensemble that can capture complex relationships and often provides strong tabular-data performance.

Additional algorithms such as K-Nearest Neighbors or Support Vector Machine may be explored if they are appropriate after preprocessing. The final model will be selected from experimental evidence rather than assumed in advance.

6.6 Model Evaluation

- Accuracy: overall proportion of correct predictions.
- Precision: among bookings predicted as cancelled, how many were actually cancelled.
- Recall: among actual cancellations, how many were correctly identified.
- F1-score: balance between precision and recall.
- Confusion Matrix: counts of correct and incorrect predictions by class.
- ROC-AUC: discrimination ability across classification thresholds.

6.7 Hyperparameter Tuning

Promising models will be tuned using a systematic method such as Grid Search or Randomized Search with cross-validation. Examples of parameters that may be investigated include tree depth, number of trees, minimum samples per split or leaf, learning rate and number of boosting stages.

Tuned models will be compared with their baseline versions. The final model will be selected based on validation/generalisation performance, class-level behaviour, interpretability and overfitting considerations.

6.8 Data Leakage Prevention

Data leakage prevention is essential because the system is intended to predict cancellation before the final outcome is known. Outcome-related variables such as `reservation_status` and `reservation_status_date` will be excluded from predictive inputs for this scenario. Preprocessing statistics, encoders, scalers and feature-selection decisions will be learned from development/training data rather than from the reserved test data.

6.9 Expected Final Outcome

The expected final result is a trained and evaluated classification pipeline that accepts selected booking characteristics, applies the same preprocessing used during training, and returns a clear cancellation prediction. Where supported, a probability or risk score may also be displayed.

7. Reason for Choosing the Dataset

7.1 Overall Justification

The Hotel Booking Demand dataset combines a clear real-world business problem with a structure that is suitable for a complete data-mining workflow. It contains a well-defined target, a large number of labelled records and a diverse set of numerical and categorical predictors.

7.2 Suitability for the FDM Mini Project

- Clear binary target suitable for supervised classification.
- 119,390 records provide sufficient data for training and evaluation.
- Mixed numerical and categorical features create meaningful preprocessing tasks.
- Missing data, outliers and moderate class imbalance provide realistic data-quality issues to investigate.

- Potential leakage variables provide an opportunity to demonstrate correct experimental design.
- The dataset supports feature engineering, feature selection, at least four classification models, tuning and systematic comparison.
- The final prediction problem can be implemented as an understandable frontend-backend application.
- The business scenario is easy to explain to both technical and non-technical audiences.

7.3 Practical and Viva Suitability

The problem can be explained simply: the system uses historical booking characteristics to predict whether a reservation is likely to be cancelled. This makes the relationship between the problem, target variable, model and final system easy to demonstrate.

The project also contains enough technical depth for individual viva questions about missing values, encoding, class imbalance, leakage, model selection, evaluation metrics, tuning and feature importance.

8. Data-Quality Observations and Limitations

8.1 Preliminary Data-Quality Observations

Known missing values occur in variables including children, country, agent and company. Their extent and meaning will be examined before choosing an imputation or exclusion strategy.

Duplicate records will be investigated. Numerical fields such as ADR, lead time, stay duration and guest counts will be checked for unusual or extreme values.

8.2 Feature Quality and Timing

Not all available columns are automatically suitable predictors. The group will define the intended prediction point and retain only information that would realistically be available at that point. This is especially important for reservation-status fields and other variables that may be updated after the original booking.

8.3 Class-Related Issues

The 63:37 target distribution is moderately imbalanced. Stratified splitting and class-sensitive metrics will be used to assess whether the model performs adequately on cancelled bookings.

8.4 Important Dataset Limitations

- The data represents only two hotels, so findings may not generalise to all hotel businesses.
- The arrival period is historical (2015-2017), so booking behaviour may differ in later years.
- Country, market and distribution patterns may differ across regions and hotel types.
- Some fields may be operationally updated after booking, requiring careful timing analysis.
- Associations discovered in observational booking data should not be interpreted automatically as causal relationships.

8.5 Proposed Data-Quality Handling Approach

After approval, the group will document dataset dimensions, data types, missingness, duplicates, unusual values, category frequencies, target distribution and potential leakage. Each preprocessing decision will be justified based on observed evidence rather than applying transformations automatically.

9. Ethical, Privacy, Licensing and Accessibility Considerations

9.1 Privacy and Data Use

The project will use the publicly available academic dataset for educational purposes. The final system should request only the information required for prediction and should not add unnecessary personal identifiers.

9.2 Responsible Use

The model's prediction should be treated as decision-support information. A predicted cancellation risk should not be used as the sole basis for unfair customer treatment, denial of service or punitive action.

9.3 Fairness and Bias

Model behaviour may differ across hotel types, market segments, countries or customer groups. Where appropriate, the group will inspect class-level and subgroup patterns and communicate limitations rather than claiming universal performance.

9.4 Transparency and Explainability

The project will document the selected features, preprocessing steps, algorithms, tuning process, evaluation metrics and limitations. Depending on the final model, feature importance, coefficients or other interpretation methods may be used to explain which variables influence predictions.

9.5 Licensing and Attribution

The dataset will be attributed to the Kaggle distribution and the original Hotel Booking Demand research publication. The group will follow the access and reuse conditions shown by the source used for the project and will not present the dataset as data collected by the student group.

9.6 Accessibility of the Final Interface

- Use clear labels and understandable descriptions for input fields.
- Provide validation messages for missing or invalid values.
- Use readable text for the final prediction and probability/risk information.
- Avoid relying on colour alone to communicate important outcomes.
- Use simple language so that a non-technical user can understand the required inputs and result.

10. Conclusion and References

10.1 Conclusion

The Hotel Booking Demand dataset is suitable for the IT3051 Fundamentals of Data Mining Mini Project because it provides a clear real-world binary classification problem, 119,390 labelled booking records, mixed feature types and sufficient complexity for EDA, preprocessing, feature engineering, leakage prevention, model comparison and optimization.

The proposed Hotel Booking Cancellation Prediction System will predict whether a booking is likely to be cancelled or not cancelled using relevant information available before the final outcome is known. The project will compare at least four classification algorithms, evaluate them with suitable metrics, tune promising models and integrate the selected pipeline into a user-facing prediction system.

The dataset and proposed problem will be submitted for lecturer validation and formal approval before proceeding to detailed EDA, preprocessing, model training or system development.

10.2 References

Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41-49. <https://doi.org/10.1016/j.dib.2018.11.126> Kaggle. Hotel Booking Demand. <https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>
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